南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

南方科技大学
物理系 Department of Physics SUSTech

$GdTaO_4$的合成及其绝热去磁制冷研究

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研究背景

极低温技术的发展及其需求

普冷 Organic Life

液氮

低温

Universe

温度（K）

10^3

10^2

10^1

10^0

10^{-1}

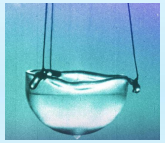
10^{-2}

冰箱、空调

超导

液化天然气、空气液化分离

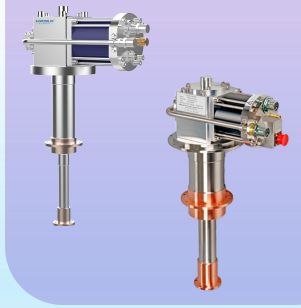
医用核磁共振、低温医疗



液氦

超流

GM、脉冲管制冷机



极低温 （亚开-毫开温区）

微波背景辐射（红外）

引力波探测（低温镜面）

深空观测（超导ESD）

极低温传感器

国家战略需求

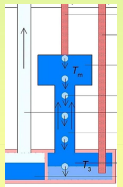
量子计算

量子精密测量

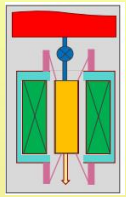
量子输运（FQHE、IQHE...）

量子磁性、超导、超流、中子散射、NMR、μSR...

前沿基础科学研究

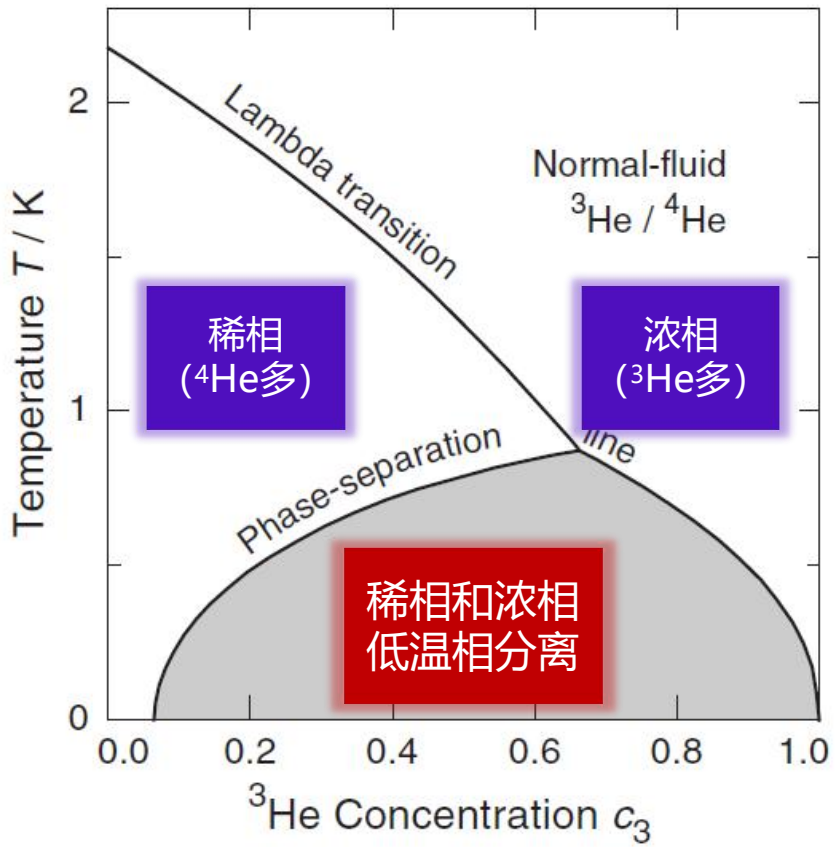


稀释制冷

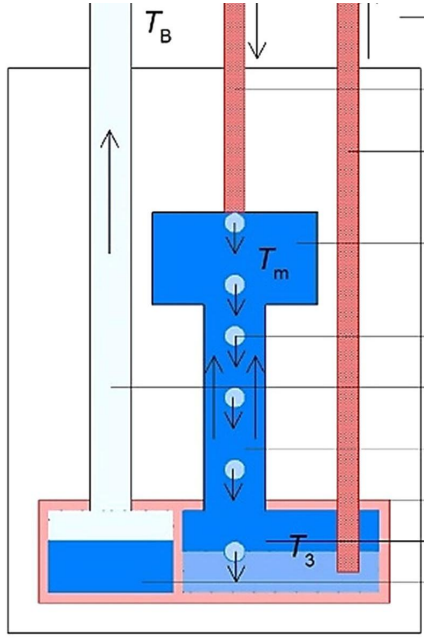


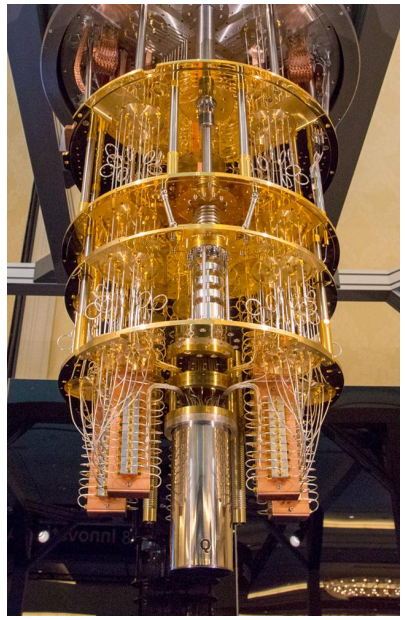
绝热去磁制冷

稀释制冷技术及其问题与挑战



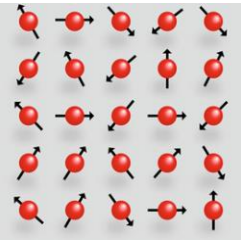
稀释制冷原理：利用^{3}He从浓相向稀相扩散吸热



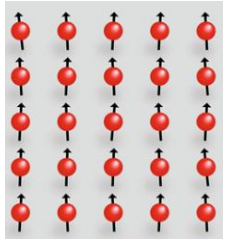


- 依赖于重力分相，空间站、深空探测等微重力环境使用受严重限制
- ^{3}He同位素：高度稀缺战略资源，核武器氚衰变的副产品（美国、俄罗斯）

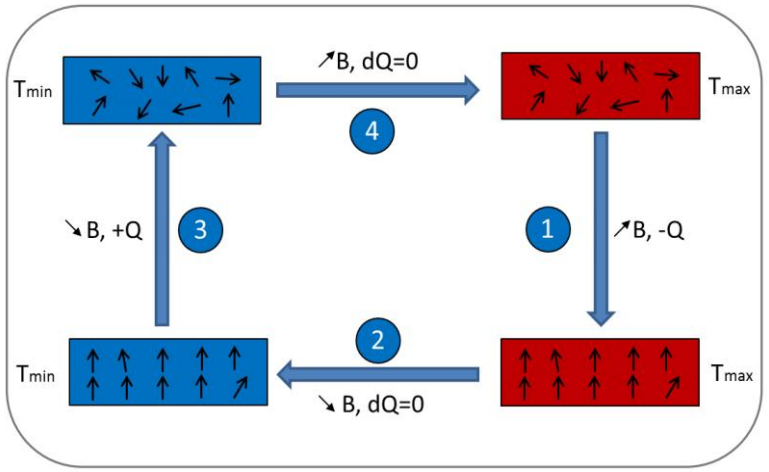
绝热去磁制冷技术原理



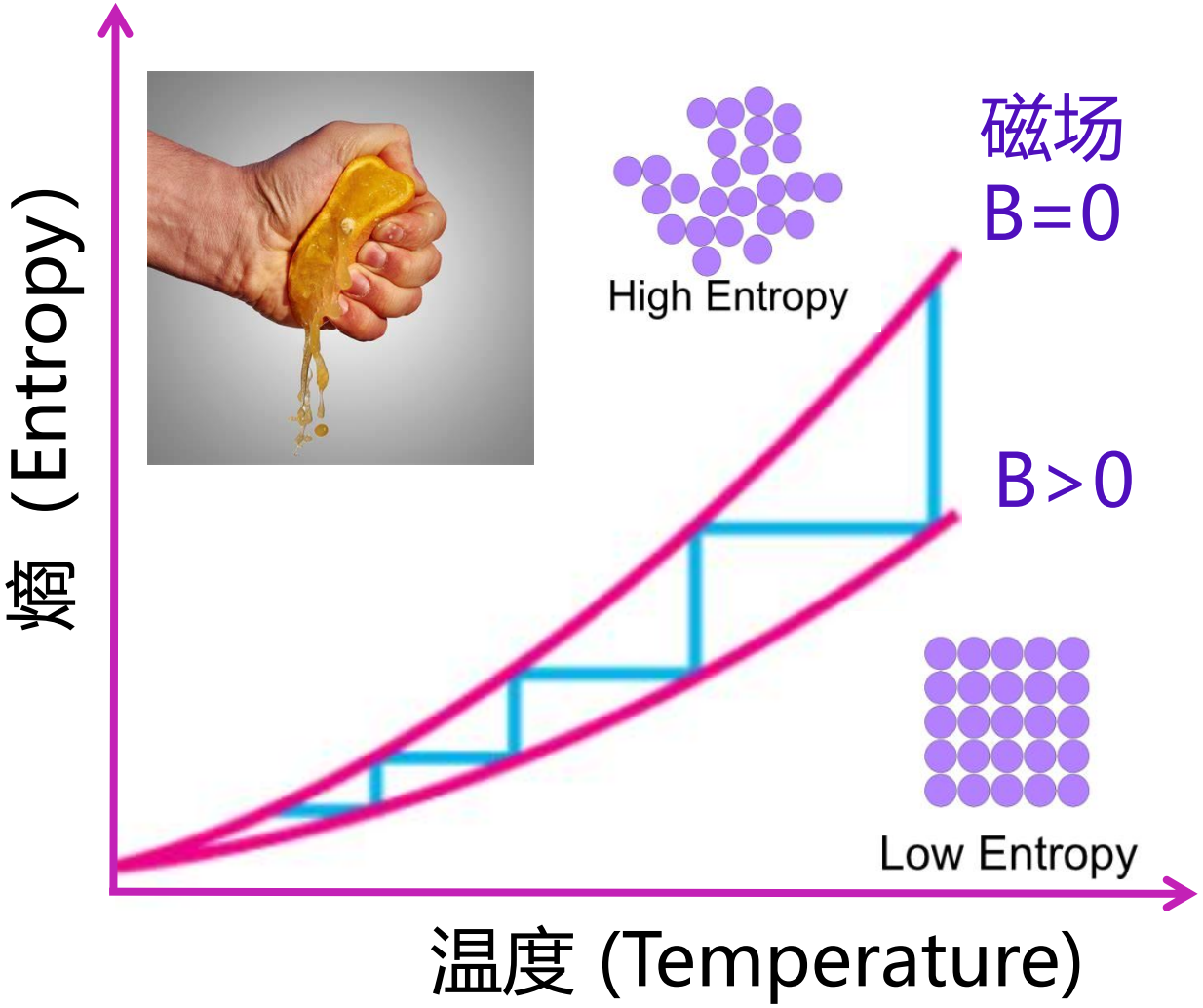
B=0
无序，磁熵大



B>0
有序，磁熵小



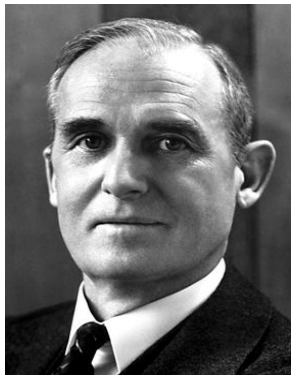
理想顺磁材料+卡诺循环



绝热去磁制冷的过程：利用磁场反复抽取制冷材料中磁熵的过程！

绝热去磁制冷技术的发展

理想顺磁体：利用结构中的**结晶水**，增加离子间距离，降低磁相互作用，保持低温无序及磁熵



诺贝尔奖
(1949)

1933年，Giauque等人
首次实现 250 mK低温

$NH_4Fe(SO_4)_2 \cdot 12H_2O$　　$KCr(SO_4)_2 \cdot 12H_2O$

目前几乎所有商用及空间任务绝热去磁制冷机，
均选用这两种晶体
(磁离子密度低， 相互作用小)

Astro-H, 2017
(and Astro-E2, 2012)

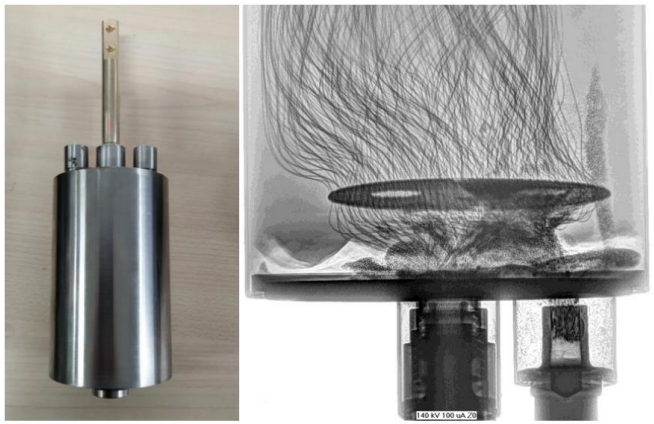
X-ray
Astronomy

50 mK

ADR

绝热去磁制冷技术的问题与挑战

- 结晶水在极低温、高真空环境下易丢失（需要特殊的封装工艺）
- 水合盐热导率低，冷量难以导出（昂贵且复杂的生长工艺，晶体需生长在金丝上）



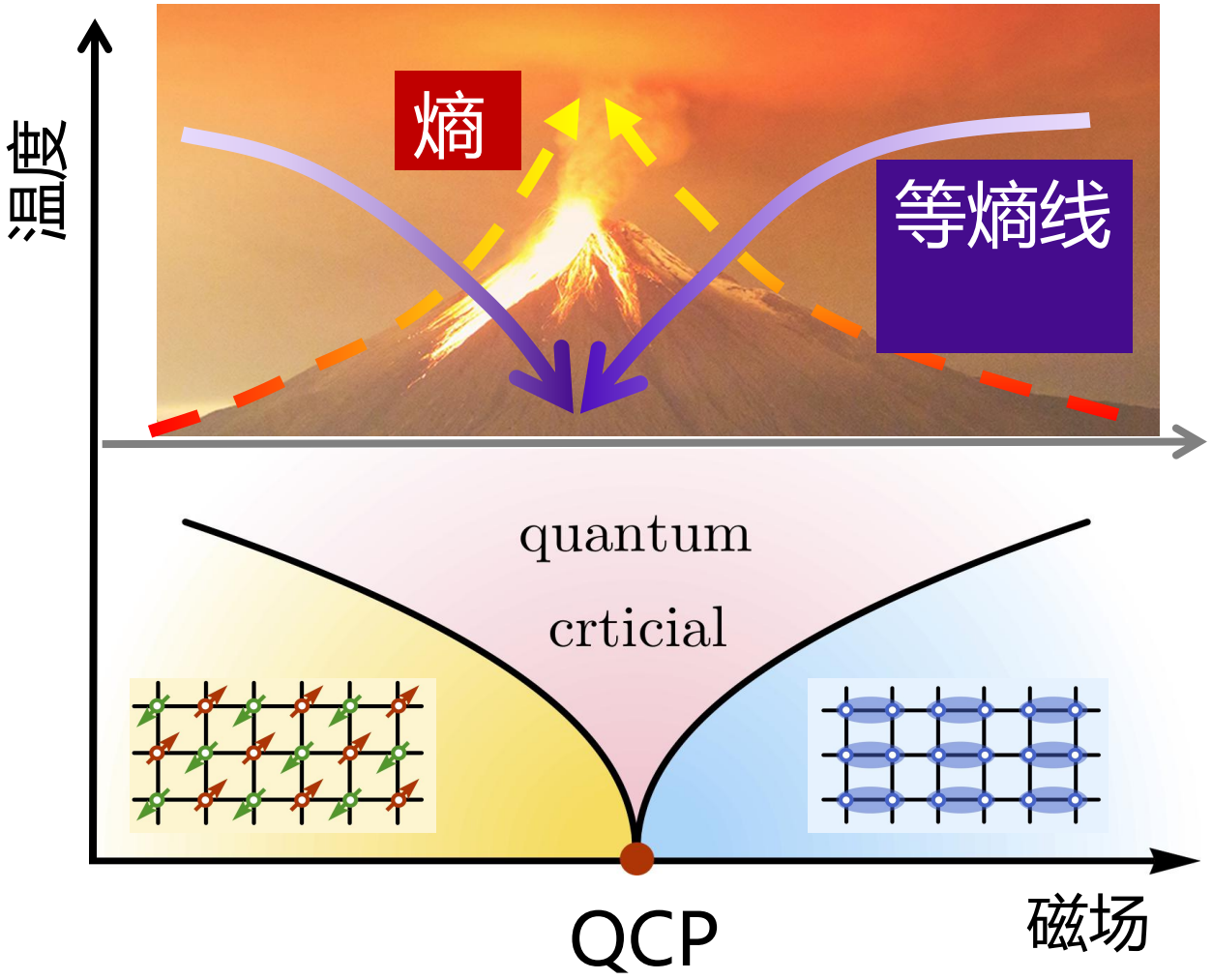
H. Jin, et al, *Ultraviolet to Gamma Ray*, (2020)

问题与挑战

生长封装技术工艺已经形成专利壁垒， 而根源在结晶水

关键：寻找稳定的、不含结晶水、高热导率、高磁离子密度的新材料！

量子临界的磁热效应



量子临界点：磁熵发散、极大的临界磁热效应

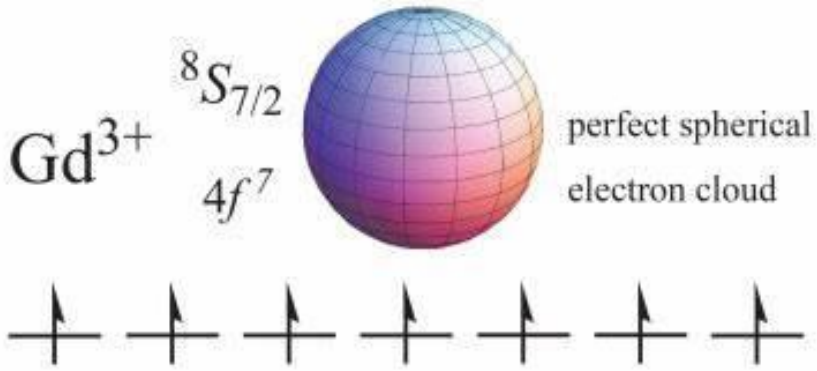
研究目标

利用低维阻挫量子磁性材料的临界磁热效应，实现极低温制冷及应用

- 磁离子密度高
- 结构稳定，无结晶水
- 低温热导率高

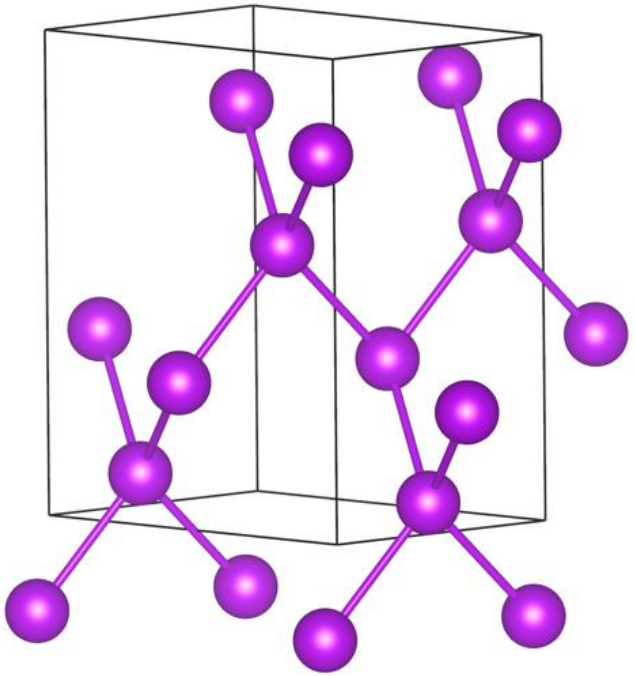
完美避开传统含水顺磁盐的困境！

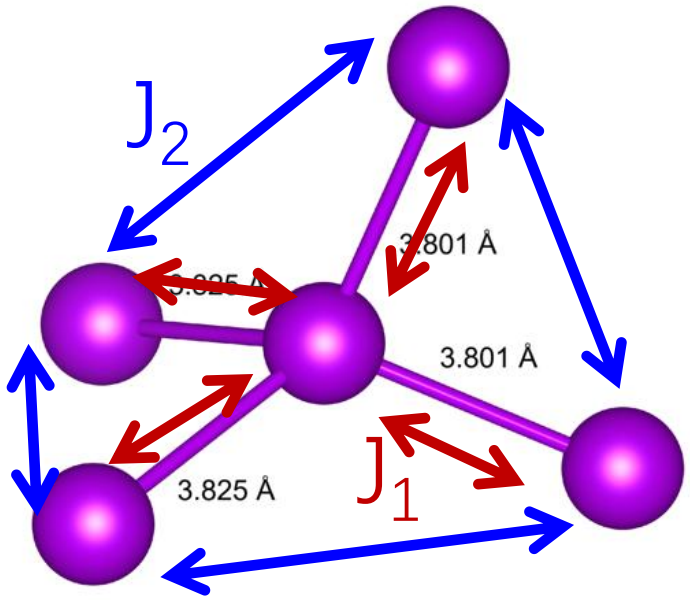
$GdTaO_4$：晶体结构



(S=7/2，L=0， J=7/2)

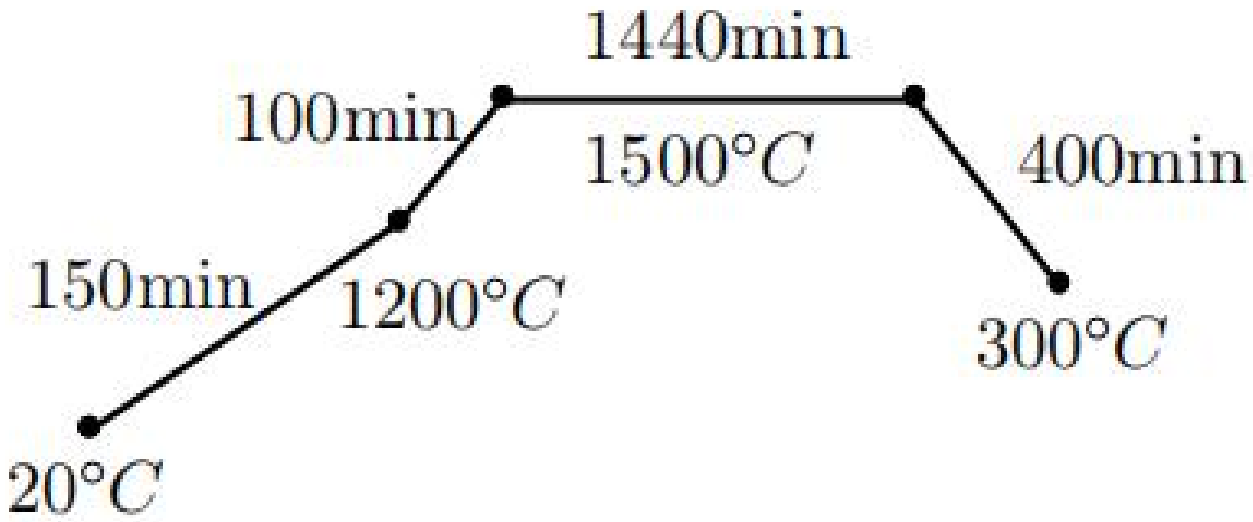
电子云：完美球型、各向同性

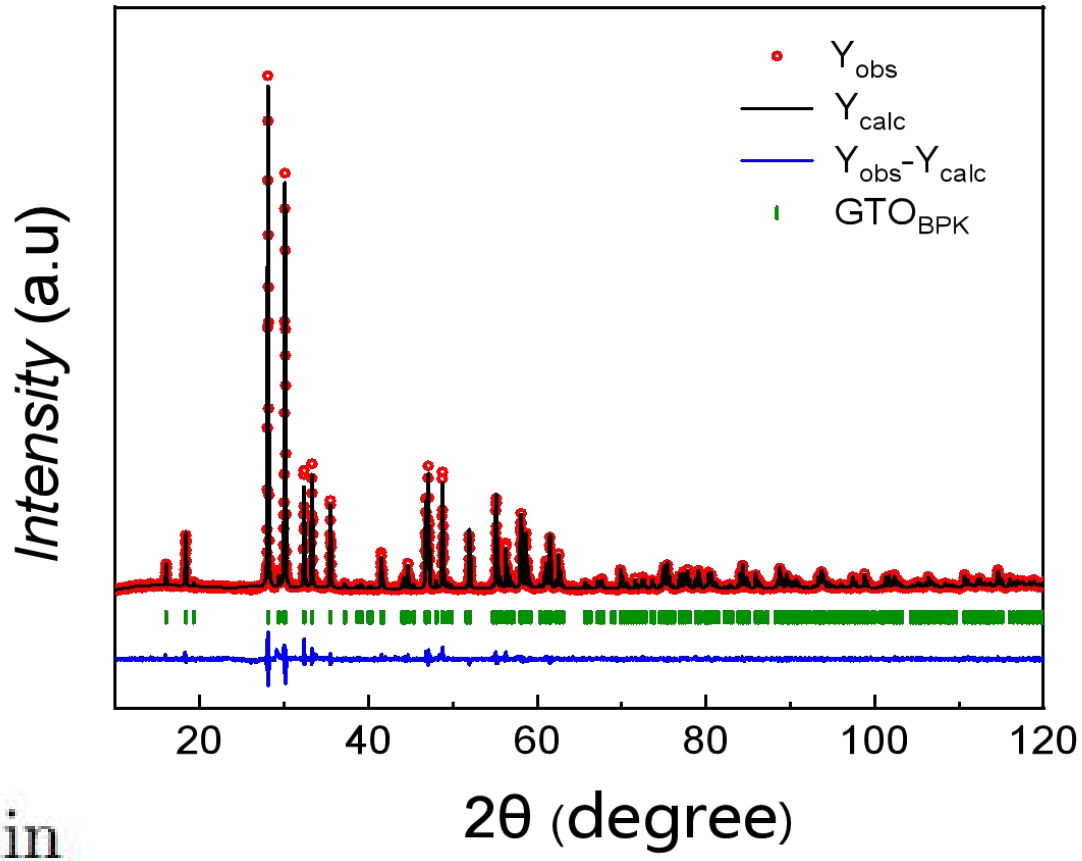




- 畸变的金刚石晶格体系：自旋阻挫、强量子自旋涨落
- 高自旋（S=7/2）、高离子密度、稳定、易于合成

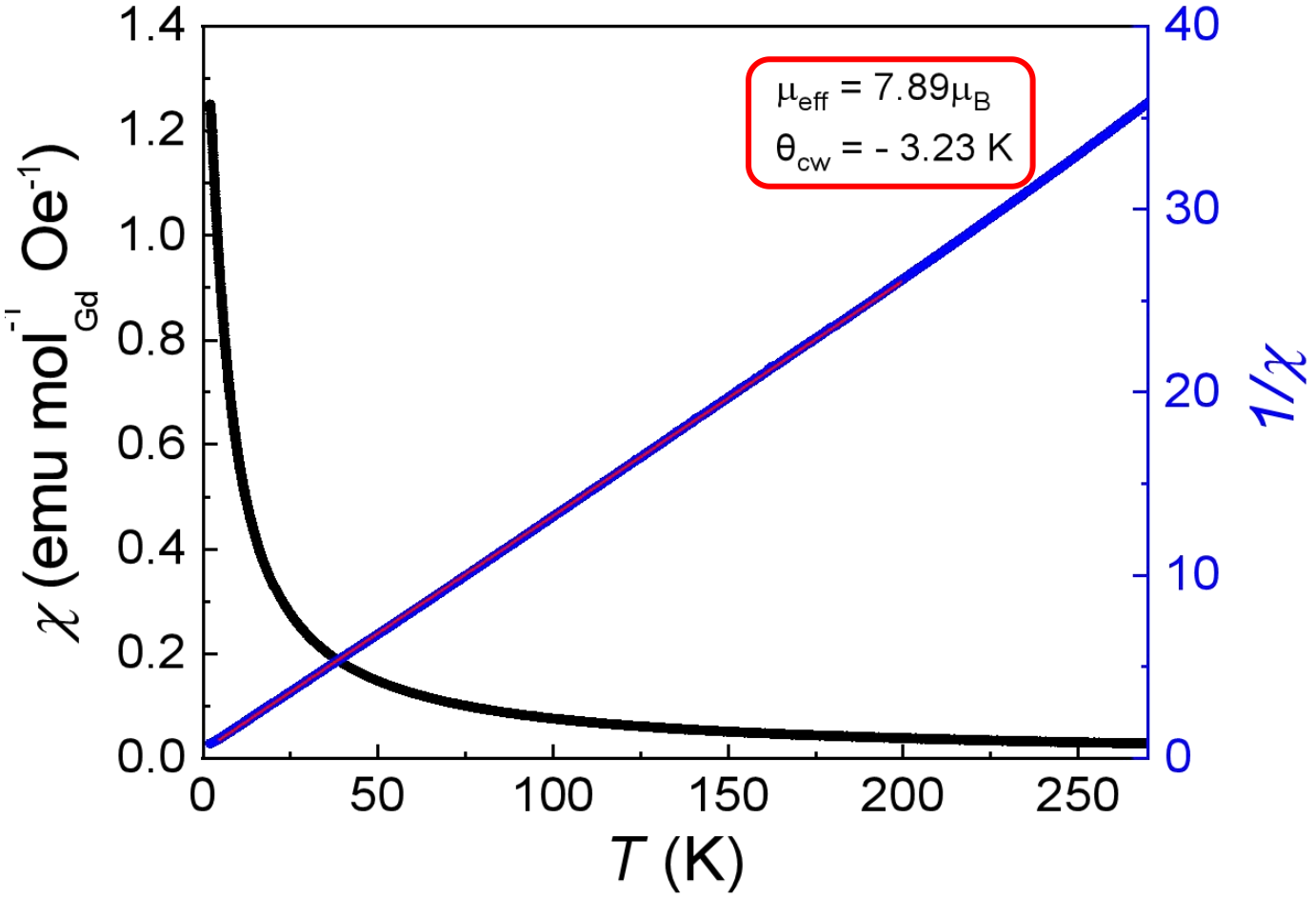
固相反应法制备$GdTaO_4$样品





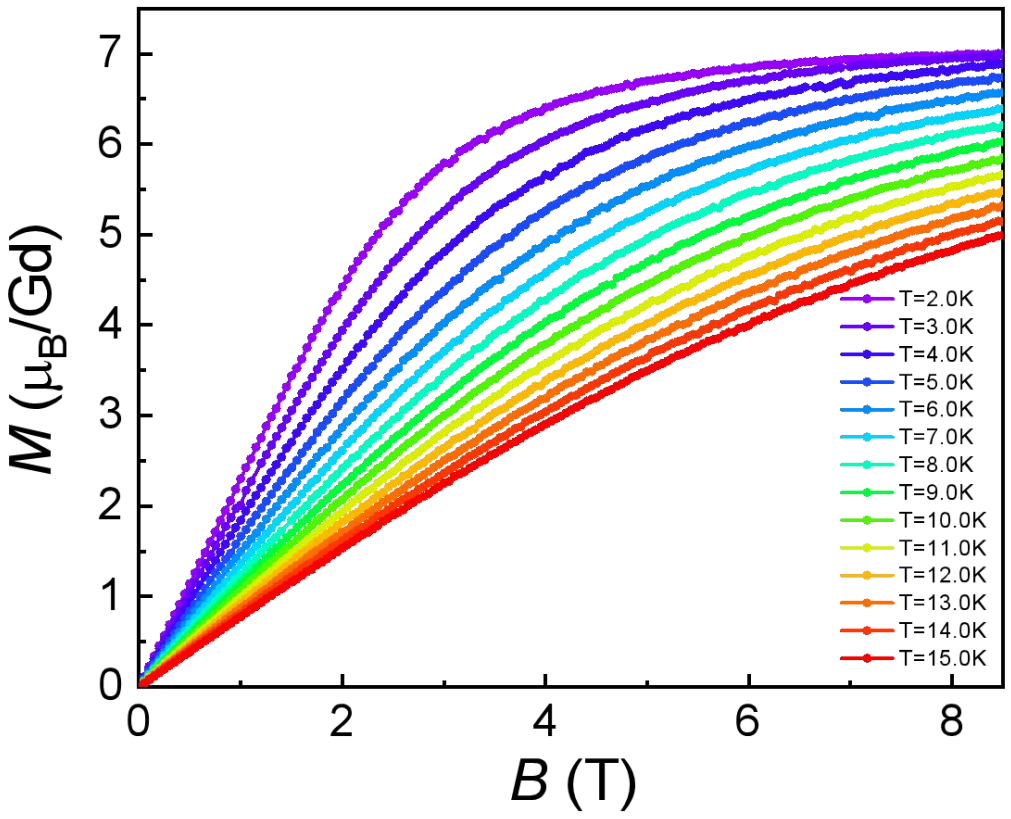
- 熔点高：1500度高温烧结
- 纯相多晶，无杂质、一致熔融、稳定

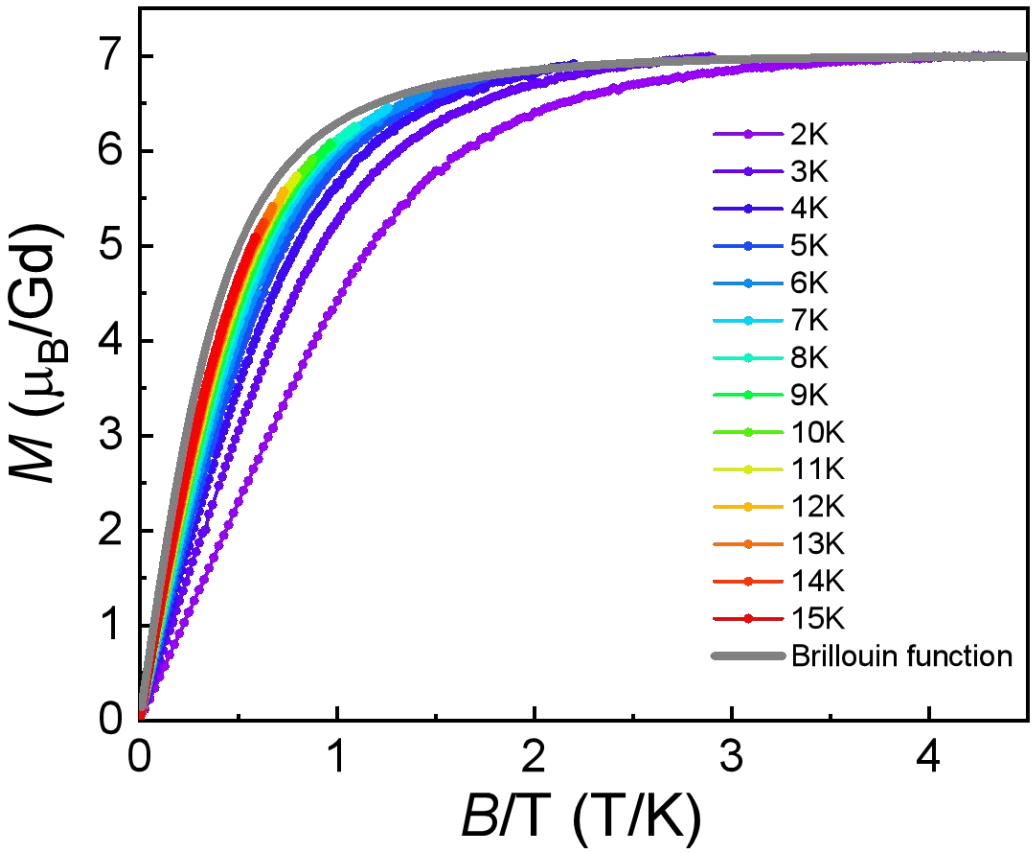
$GdTaO_4$ 磁化测量及分析



- 反铁磁相互作用、没有杂相、2K之上没有相变

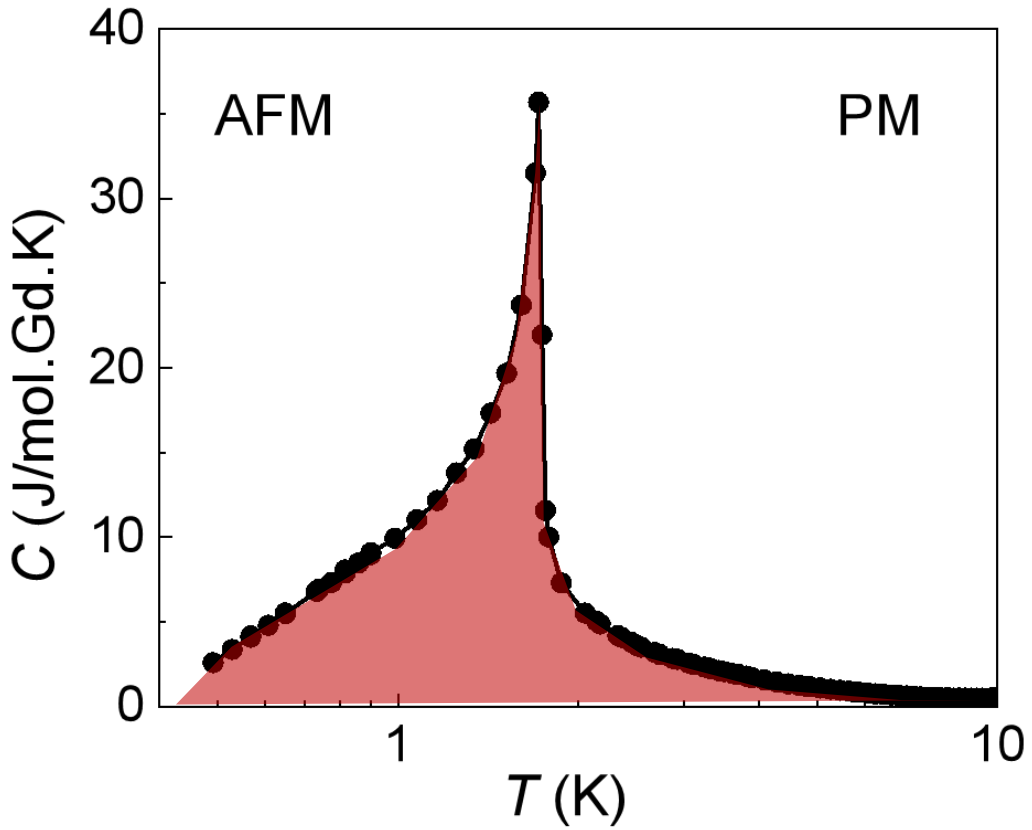
$GdTaO_4$ 磁化测量及分析

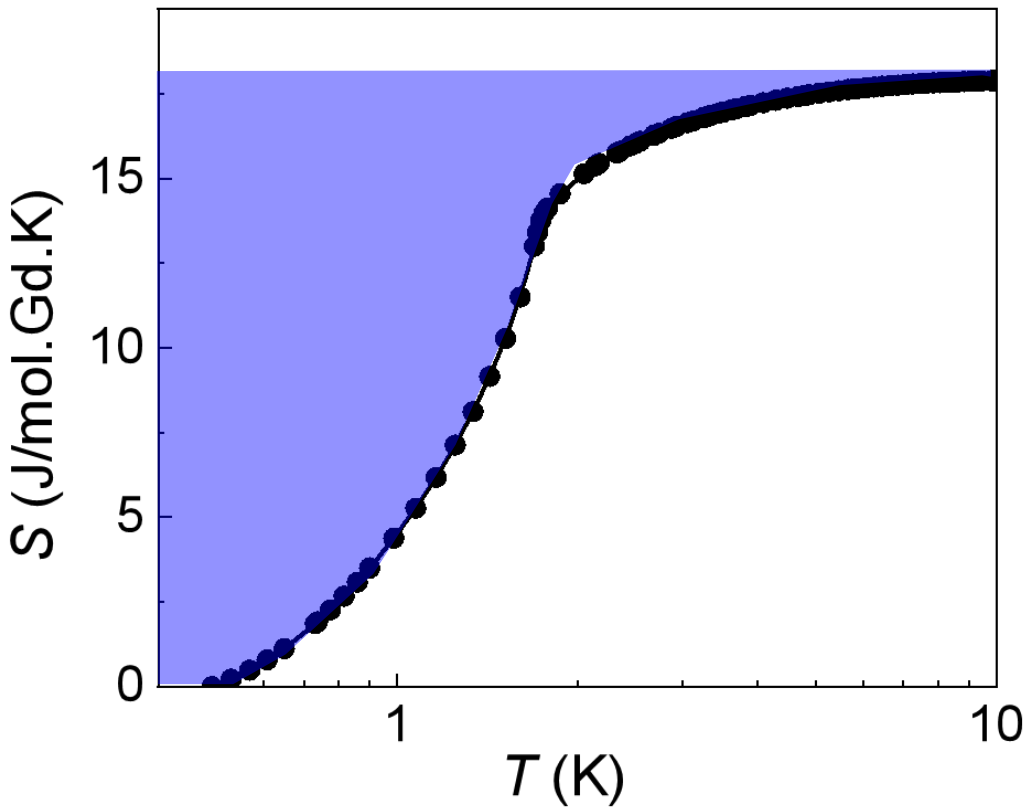




- 反铁磁相互作用、低温偏离顺磁行为（偏离布里渊函数）

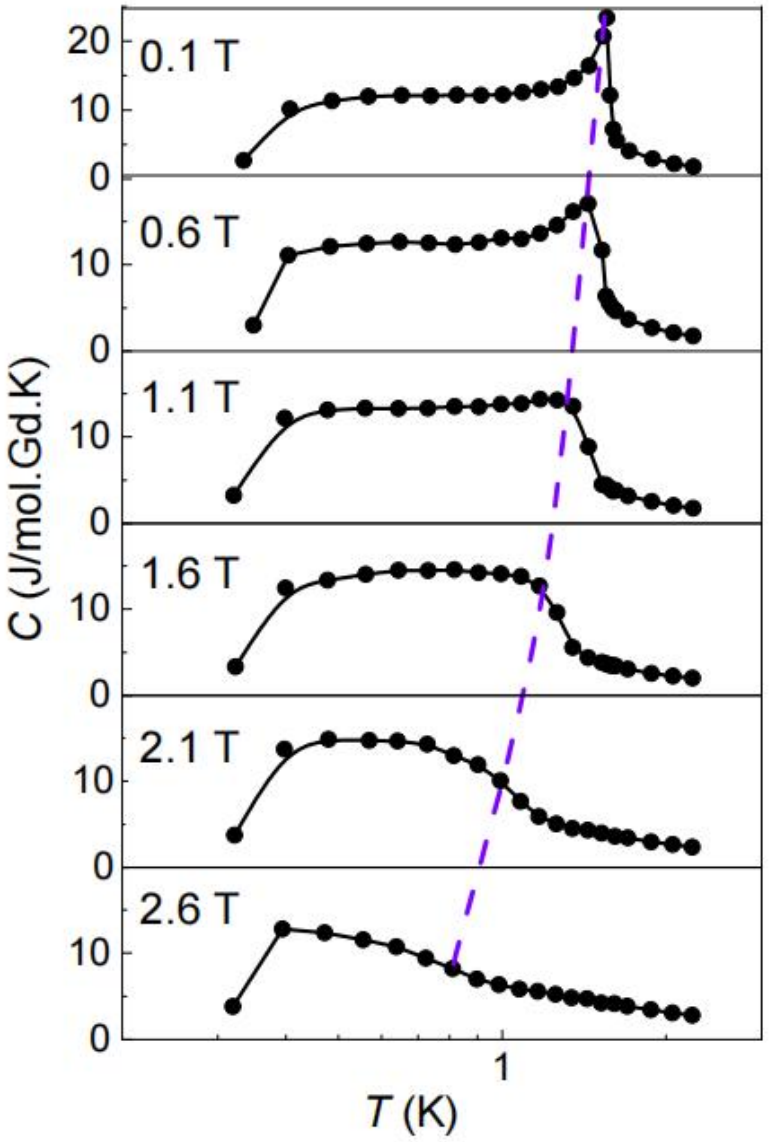
$GdTaO_4$ 零场比热测量及分析

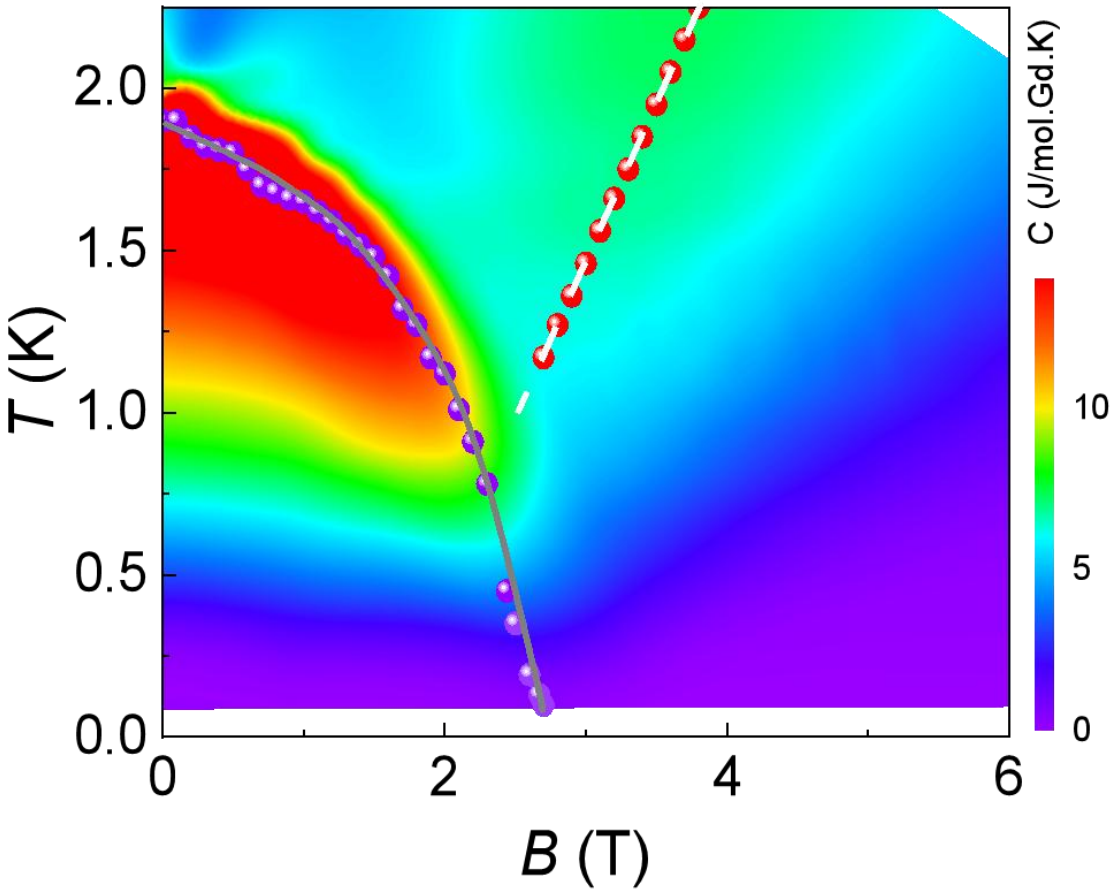




- 低温进入反铁磁： T_N=1.7K
- 1-2K温区：巨大的磁比热贡献，兼具蓄冷功能

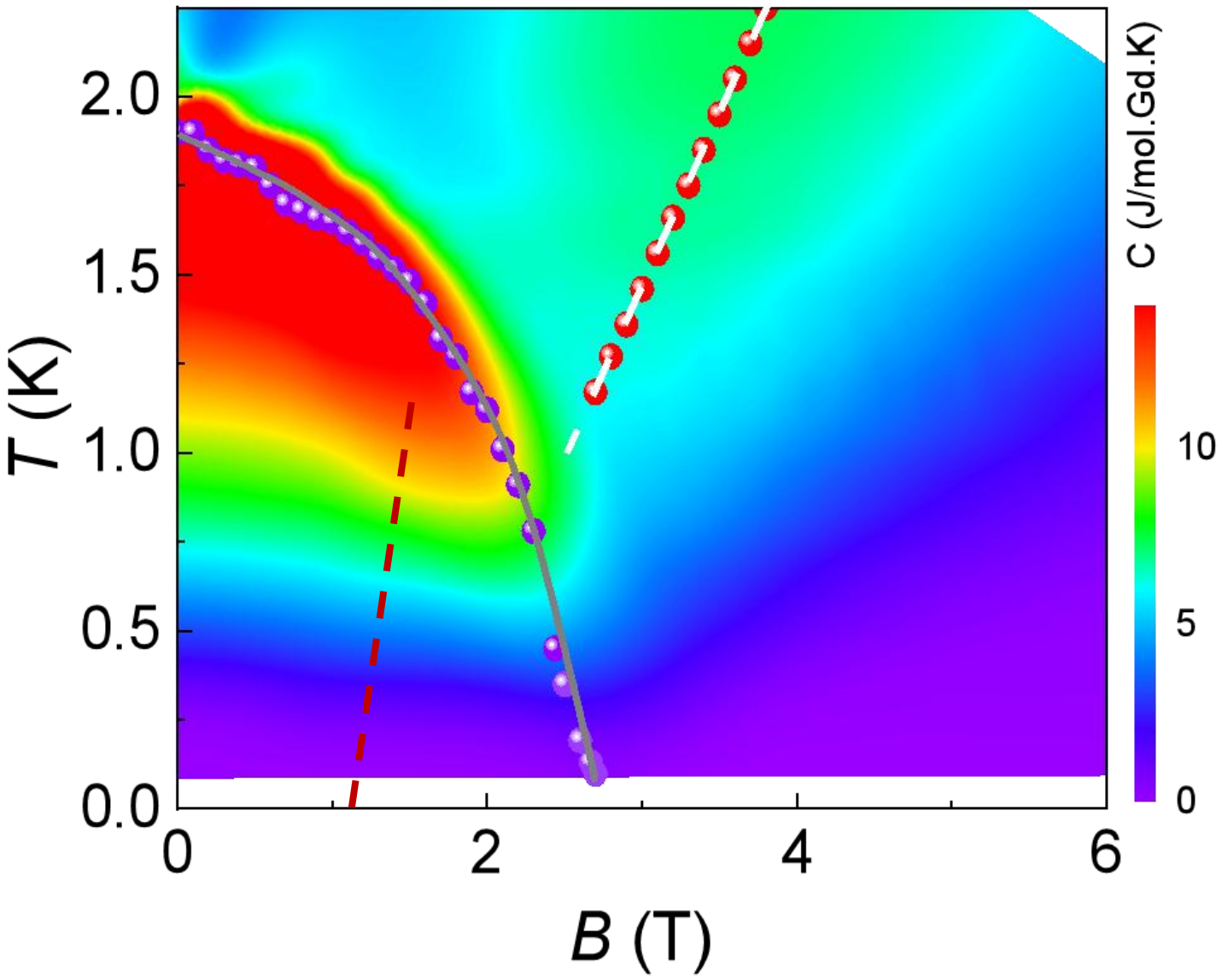
$GdTaO_4$ 比热及相图

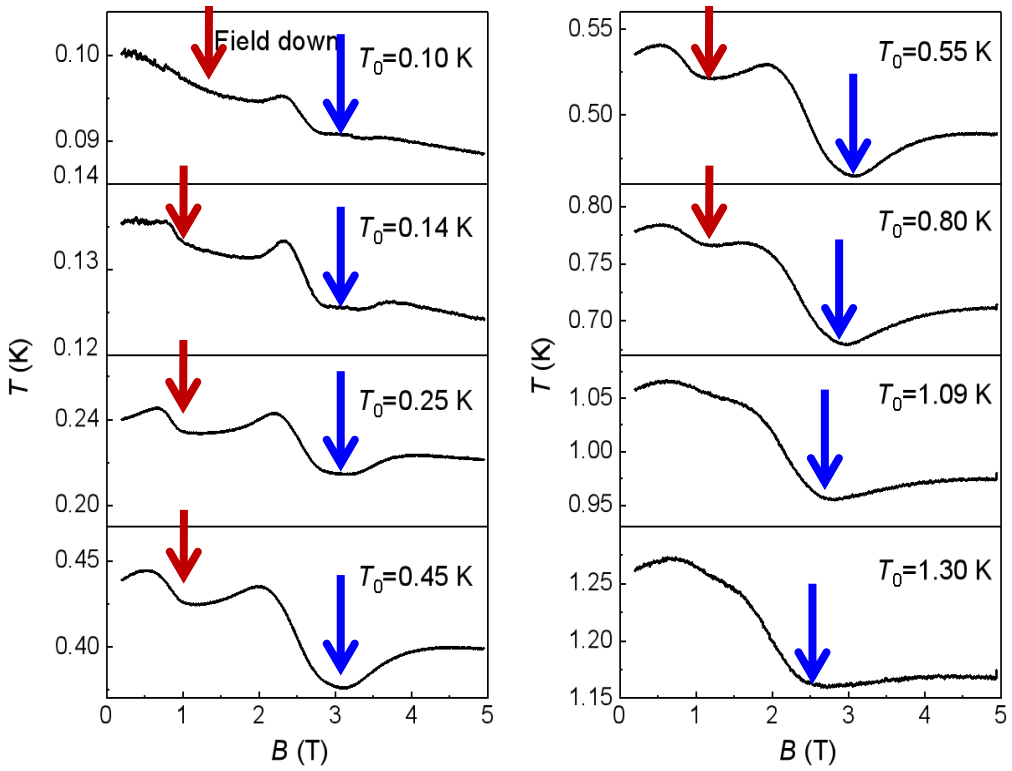




- 磁场增加，磁有序被压制
- 量子临界磁场：B_c = 2.9 T

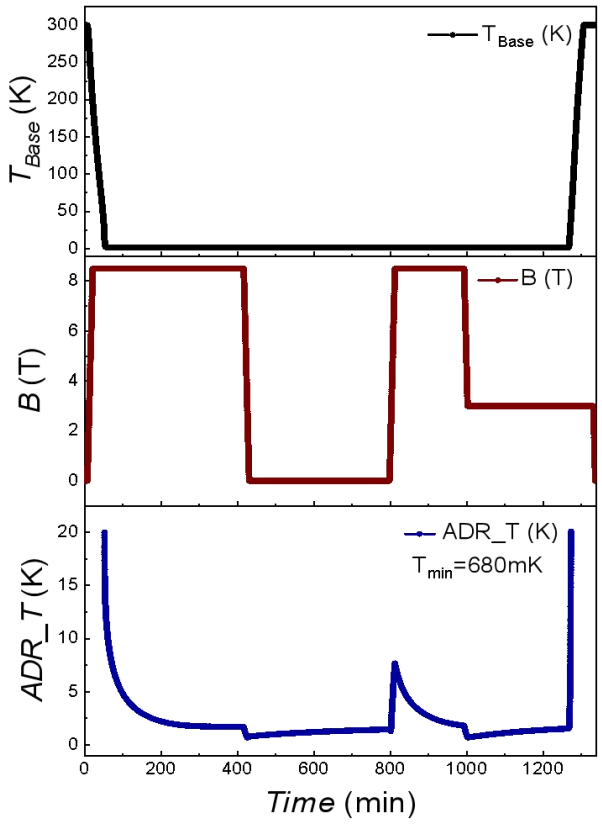
$GdTaO_4$ 相图及磁热效应

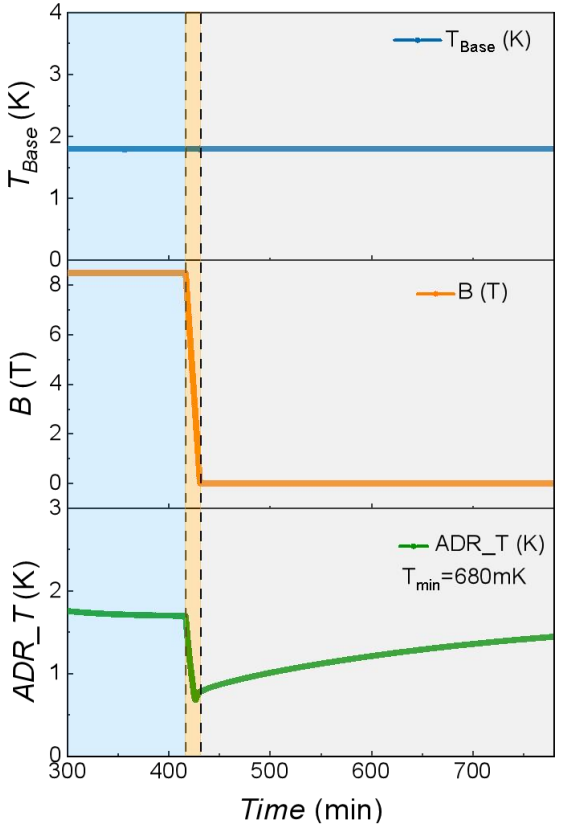


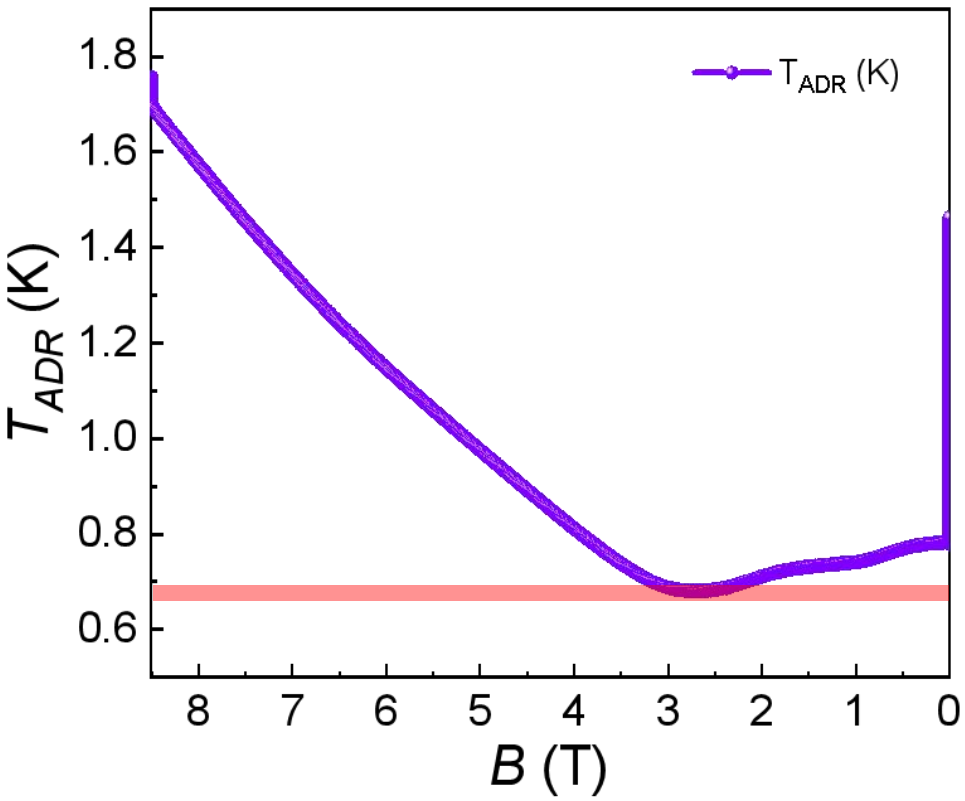


- 极小样品（0.58mg，高真空）
- 磁场诱导相 （future work）
- 140mK温区有明显制冷效果

$GdTaO_4$ 绝热去磁制冷应用





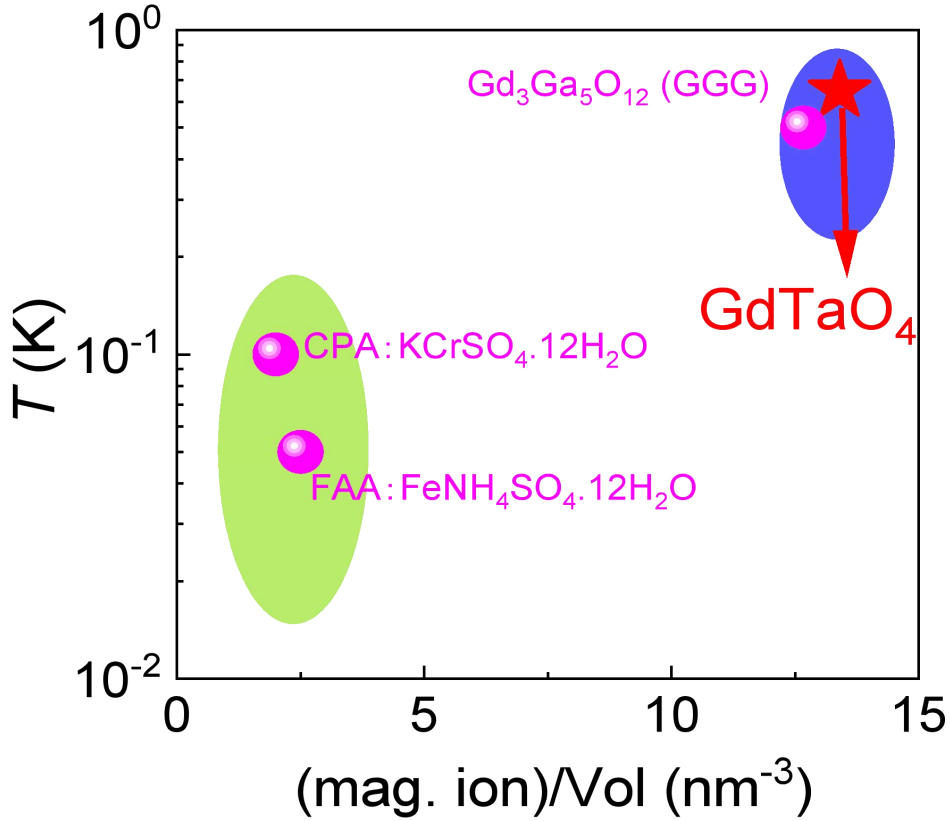


15g 多晶样品，节约测试等待时间，粗真空 （10 torr），加快降温速度

退磁后，样品在1.5K以下保持超8小时，最低制冷温度达680mK。

$GdTaO_4$ 绝热去磁制冷应用

QD开发的基于CPA晶体的商业化绝热制冷器件在350K之上完全失效



- 密度和制冷温度可媲美商业化GGG晶体
- 易于合成、成本低、极其稳定（熔点2200度）

未来：高真空+大单晶，减少漏热，最低制冷温度和保温时间均可进一步提升！

展望

物理层面：单晶，低温磁场诱导相的微观机理，几何阻挫

应用层面：高致密度大尺寸陶瓷、高真空、实际制冷应用

感谢各位老师百忙之中莅临指导，

请批评指正！